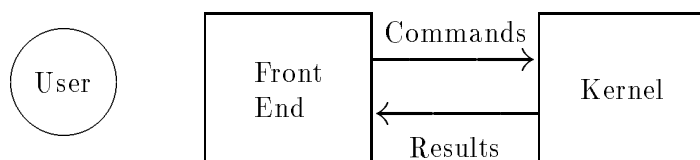


■ 1.3 Using the *Mathematica* System

■ 1.3.1 *Mathematica* Front Ends

There are really two parts to the *Mathematica* system: a “kernel” that actually does computations, and a “front end” which manages interaction with the user.

When you type in $2 + 2$, it is the kernel that computes the result 4. The front end is responsible for reading your input, and displaying the final result.



Most of this book is about what the kernel of *Mathematica* can do. This section, however, discusses some aspects of *Mathematica* front ends.

The kernel of *Mathematica* works in the same way on all computers that run *Mathematica*. This means for example that any of the dialogs with *Mathematica* which you have seen so far would run exactly the same on any kind of computer.

Although the kernel works the same, there can be very different front ends for *Mathematica* on different computers. The basic idea is for the front end to take advantage of whatever graphical or other capabilities each particular kind of computer has.

For information about the front end for *Mathematica* on your particular computer, you should look at the documentation that came with your system. This section discusses generalities, and does not give all the details you need to make use of a particular *Mathematica* front end.

Line-by-line (e.g. printing terminal).

Full screen text (e.g. standard display terminal).

Graphical (e.g. Macintosh).

Three types of *Mathematica* front ends.

The dialogs in this book essentially correspond to what you get by using *Mathematica* with a line-by-line front end. *Mathematica* prints a “prompt” of the form

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

© 1988 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.

`In[n] :=`. You type your input, and *Mathematica* responds by printing out a result.

On many modern personal computers and workstations, however, there are *Mathematica* front ends which make more sophisticated interactions possible. Different front ends in general have different capabilities, but there are a number of typical features.

The first thing that more sophisticated front ends typically give you is the ability to browse through all the previous lines in your *Mathematica* session. When you are trying to work what input to give next, you will often find it useful to “scroll backwards”, and look through previous results that you got. If you have any kind of full-screen display, you should be able to get a *Mathematica* front end which allows you to do this kind of thing.

Browse through your *Mathematica* session.

Work with *Mathematica* from several different windows.

Have *Mathematica* find possible completions for parts of command names you type.

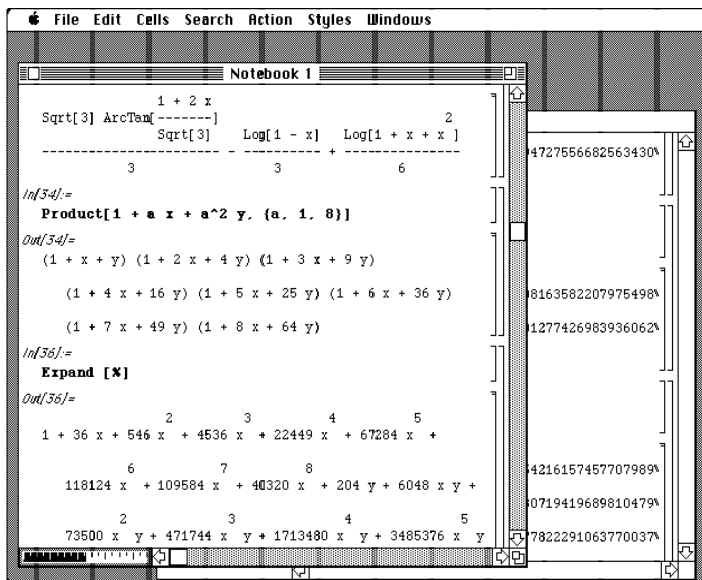
Search for text in your *Mathematica* session.

Copy and paste text and *Mathematica* expressions.

Some things you can do with more sophisticated front ends.

Here is a picture of the screen of a Macintosh computer that is running *Mathematica*.

There are two *Mathematica* windows on the screen. On the right-hand side of each window is a “scroll bar”. You can move the scroll bar with the mouse, to look at different sections of your *Mathematica* session.



Although different front ends for *Mathematica* may look very different, it is important to remember that the actual commands you give to do calculations in *Mathematica* are the same in all cases. The only thing that changes is exactly how you construct and enter the commands, and how the results from them are displayed.

Advanced *Mathematica* front ends can save you a lot of typing. Instead of having to enter each line of input explicitly, you can often construct your input by pasting together pieces of text that you pick up from different places using graphical tools. The pieces of text may come from input that you gave earlier in your session, or from output that *Mathematica* generated.

Another important feature of advanced *Mathematica* front ends is that they typically give you a number of ways to organize the results you get from *Mathematica* calculations. Many front ends for example allow you to use multiple windows for *Mathematica* input and output. You can then use different windows for different parts of your calculation.